HSIANGKUN “SHAWN” SHIH

QC CHEMIST | ANALYTICAL CHEMISTRY | BIOCHEMISTRY RESEARCH

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PROFESSIONAL PROFILE

ACS-certified chemistry graduate and Quality Control Chemist with experience in regulated chemical manufacturing, quantitative testing, spectroscopy, data integrity, and undergraduate chemical-biology research. Brings hands-on work in titration, standards and sample preparation, UV-Vis and fluorescence analysis, SOP/GMP/GDP documentation, OOS investigation support, and research communication. Interested in chemistry, QC, analytical development, biotech, and research roles where careful experimentation and reproducible results matter.

PROFESSIONAL EXPERIENCE

Quality Control Chemist | Ricca Chemical Company | *Jul 2025–Nov 6, 2026*

Pocomoke City, Maryland

- Perform titrations and quantitative analyses to support batch-release testing of chemical products and laboratory reagents.
- Operate UV-Vis and fluorescence instrumentation for analytical sample characterization; use calibrated pH and conductivity meters for routine testing.
- Prepare standards, dilutions, buffers, and samples while maintaining chain-of-custody, equipment-use, and test records.
- Document work in accordance with GMP, GDP, and approved SOPs; maintain accurate, review-ready entries in LIMS and controlled records.
- Investigate out-of-specification and atypical results and support assay or method-validation activities by documenting observations and quality-impacting variables.

AI Trainer | Handshake | *Nov 2025–Present*

Remote

- Apply structured analytical review to scientific and technical outputs, developing reproducible validation and triage workflows that improve consistency and documentation quality.

Program Coordinator | CCACC Academy | *2019–2024*

Rockville, Maryland

- Coordinated program logistics, maintained safety and operational documentation, and mentored students through hands-on STEM and chemistry outreach activities.

Service Operations Associate | Ebisu Lifestyle / Hardee’s | *2024*

Delmar, Delaware

- Supported SOP-driven operations, sanitation compliance, teamwork, and reliable execution in a fast-paced service environment.

RESEARCH EXPERIENCE

Undergraduate Researcher | Salisbury University, Department of Chemistry | *2023–2025*
Faculty Mentor: Joshua Sokoloski
Project: Targeting Cancer-Associated Proteins and Click Chemistry Labeling of Probes for Tumor Imaging and Ablation

- Developed and evaluated copper-catalyzed azide–alkyne cycloaddition (CuAAC) strategies to label DNA and RNA probes with fluorophores and explore attachment of iron(II)-based MRI contrast agents.
- Used bioinformatics and nucleic-acid sequence analysis to identify accessible target regions and evaluate sequence-dependent copper interactions.
- Prepared reaction mixtures, optimized incubation and solvent conditions, and purified labeled products using size-exclusion spin columns and dialysis.
- Quantified labeling efficiency and studied nucleic-acid/metal interactions using UV-Vis and fluorescence spectroscopy, thermal-denaturation analysis, and gel electrophoresis.
- Contributed to experiments involving cancer-associated targets including MDMX, p53, and HOTAIR; synthesized findings through literature review, data analysis, posters, and public-facing scientific communication.
- Research supported in part by NIH AREA Grant 1R15EB029559-01 and the Henson Research Fellowship.

SELECTED PRESENTATIONS & SCHOLARLY COMMUNICATION

Posters on the Bay | Maryland State House / Salisbury University | *Feb 2025*
Co-presenter with Min Thway Khant, "Targeting Cancer-Associated Proteins and Click Chemistry Labeling of Probes for Tumor Imaging and Ablation." Presented undergraduate research to Maryland government and university audiences.

Salisbury University Student Research Conference | Salisbury University | *2024*
Co-presenter with Min Thway Khant, "Targeting Tumor-Associated Molecules with Nucleic Acid and Peptide Probes for Imaging and Ablation."

ASBMB Annual Meeting | American Society for Biochemistry and Molecular Biology | *2022*
Presented undergraduate research with travel support from the Dorothy Ruxton Chemistry Student Research Fund.

Shawn BioChem World | Independent Science Communication Portfolio | *2024–Present*
Publish accessible explanations of click chemistry, cancer-imaging probes, analytical chemistry, water analysis, and scientific learning at chem.hsiangkun.org.

SELECTED RESEARCH METHODS

CHEMICAL BIOLOGY
CuAAC click chemistry; probe labeling; bioconjugation; nucleic-acid sequence design

SPECTROSCOPY
UV-Vis; fluorescence; FT-IR; NMR exposure

SEPARATION & PREP
Sample preparation; dilutions; buffer preparation; spin-column purification; dialysis

CHROMATOGRAPHY
HPLC/UPLC and GC exposure; purge-and-trap concepts; analytical method awareness

MOLECULAR METHODS
Gel electrophoresis; thermal-denaturation studies; Western blot; cell and protein fractionation

DATA & QUALITY
Experimental design; calibration; quantitative analysis; documentation; literature review

EDUCATION

Bachelor of Science in Chemistry — ACS Certified | Salisbury University | *Dec 2025*
Salisbury, Maryland | GPA: 3.5
Selected Coursework: Analytical Chemistry; Instrumental Analysis; Physical Chemistry

Academic Focus: Analytical chemistry, biochemistry, molecular biology, chemical biology, and scientific research

Activities: American Society for Biochemistry and Molecular Biology (ASBMB)

TECHNICAL COMPETENCIES

ANALYTICAL & QC
Titration; quantitative analysis; standards and reagent preparation; sample preparation; calibration; batch-release testing; assay validation support; method development support; OOS/atypical-result investigation

QUALITY SYSTEMS
GMP/cGMP; GDP; SOP-controlled workflows; chain of custody; data integrity; controlled documentation; laboratory safety

SOFTWARE & DATA
LIMS; Microsoft Excel and Word; GraphPad Prism; data analysis; literature review; technical writing

INSTRUMENTATION
UV-Vis; fluorescence spectroscopy; FT-IR; pH meter; conductivity meter; HPLC/UPLC, GC, and NMR exposure

BIOCHEMISTRY & MOLECULAR
Click chemistry; DNA/RNA probe labeling; gel electrophoresis; thermal denaturation; Western blot; cell and protein fractionation; bioinformatics-assisted sequence design

LANGUAGES
English — native or bilingual proficiency; Chinese — native or bilingual proficiency

AWARDS & RECOGNITION

Core Values Achievement Award — Q1 2026 | Ricca Chemical Company | *Mar 2026*

Dorothy Ruxton Chemistry Student Research Fund Recipient | Salisbury University | *2022*
Travel funding supporting presentation of undergraduate research at the ASBMB Annual Meeting.

Henson Research Fellowship | Salisbury University | *Undergraduate Research*

CREDENTIALS & PROFESSIONAL DEVELOPMENT

- ACS Certified Degree credential, American Chemical Society — issued Dec 2025 (Credential ID 176941862).
- ACS Author Lab, ACS Division of Medicinal Chemistry — issued Mar 2026.
- ACS Reviewer Lab and ACS Reviewer Lab (Chinese), ACS Division of Medicinal Chemistry — issued Mar 2026.
- Schistosoma spp. Life Cycle Training Course, Biomedical Research Institute — Apr 2026.

PROFESSIONAL INTERESTS

Analytical chemistry • Quality control • Biopharmaceutical and biotechnology laboratories • Cancer biology • Protein analysis • Bio-orthogonal chemistry • Instrumentation • Applied research • Scientific communication